📊 DATA ANALYSIS ACTIVITY PACK · YEAR 9–10

Advanced Water Safety Data Analysis Activity Pack

Three self-contained data analysis activities using real Victorian and global drowning statistics. Each activity can be used as a lesson starter, homework task, or standalone assessment item.

Year 9–10 HPE | 3 standalone activities | Real data | VC2HP10P02 · VC2HP10P08 | ~15 min each

> **About this pack:** These three activities use real, current Victorian and global drowning data. Each activity is self-contained and includes a worked example to get you started. Activities can be used in any order. Verify current data at **lsv.com.au/what-we-do/research-advocacy** and **royallifesaving.com.au/research**.

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ACTIVITY A

Trend Analysis — Victorian Drowning Rates Over 5 Years

~15 min | VC2HP10P08 | Individual | Source: lsv.com.au

The table below shows Victorian drowning death totals from 2019–20 to 2023–24. Use this data to calculate rates, identify trends, and interpret the implications.

Financial Year	Fatal Drownings (Victoria)	Vic. Population (approx.)	Rate per 100,000 (calculate)	Above/Below 5-yr average?
2019–20	46	6.63M	_____	_____
2020–21	42	6.65M	_____	_____
2021–22	47	6.68M	_____	_____
2022–23	50	6.80M	_____	_____
2023–24	54	6.90M	_____	_____
5-year average	_____	—	_____	—

Worked example (2019–20):
Rate = fatal drownings ÷ (population ÷ 100,000)
= 46 ÷ (6,630,000 ÷ 100,000) = 46 ÷ 66.3 = **0.69 per 100,000**
Tip: divide population by 100,000 first, then divide deaths by that number.

Question A.1 — Trend Description

2 marks

Describe the trend in Victorian drowning deaths from 2019–20 to 2023–24. Is the trend increasing, decreasing, or fluctuating? Use data from the table to support your description.

Question A.2 — Trend Interpretation

3 marks

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Identify ONE factor that might explain why Victorian drowning deaths increased from 2020–21 to 2023–24. Your explanation should refer to a specific social, environmental, or policy factor — not just "more people went swimming."

Source or evidence to support your explanation: _____

Question A.3 — Graph Task

3 marks

In the space below, construct a line graph showing Victorian drowning rates per 100,000 from 2019–20 to 2023–24. Label both axes, include a descriptive title, and add a horizontal line showing the 5-year average rate.

📈 *Draw your line graph here — or attach on a separate page. Include: title, labelled axes, data points connected by a line, and a dashed horizontal line for the average.*

Graph title: _____

2-sentence interpretation: _____

_____

Question A.4 — Prediction

2 marks

Based on the trend you identified, predict Victoria's approximate drowning rate in 2025–26 if no new prevention strategies are implemented. Justify your prediction with reference to the trend data.

My prediction: approximately _____ per 100,000 (_____ deaths)

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ACTIVITY B

CALD Over-Representation — Structured Argument Analysis

~15 min | VC2HP10P02 | Individual / Pairs | Source: lsv.com.au

Use the data below to analyse the over-representation of multicultural communities in Victorian drowning deaths. You will work out a share, confront what the dataset does not capture, and construct a structured argument.

Group	Deaths since 2015–16	Country of birth	Share of all deaths
People from multicultural backgrounds	185	Recorded	_____
Cases with no country of birth recorded	Not stated	Unknown	_____

Worked example — the share, and its limits:
Share = 185 deaths ÷ all Victorian drowning deaths since 2015–16 = **37%**
Interpretation: Life Saving Victoria reports that 185 people from multicultural backgrounds fatally drowned in Victoria since 2015–16 — 37% of all drowning deaths — with country of birth unknown in 22% of cases. Nationally, RLSSA reports 39% of people who drowned were born overseas, based on the 61% of cases where country of birth was known (National Drowning Report 2026). Both figures are lower bounds: the unknown share means the real burden is likely higher.

Question B.1 — Calculate & Interpret

2 marks

Calculate the over-representation ratio for non-CALD Victorians. What does a ratio below 1.0 tell you about non-CALD drowning risk relative to their population share?

Non-CALD ratio: 60 ÷ 72 = _____

A ratio below 1.0 means: _____

Question B.2 — Structural vs Individual Argument

4 marks

Construct TWO arguments explaining the CALD over-representation: one individual-level and one structural-level. Then explain which is more useful for developing prevention policy, and why.

INDIVIDUAL-LEVEL ARGUMENT (BLAMES PERSONAL CHOICES)

Claim:

Because:

Therefore:

STRUCTURAL-LEVEL ARGUMENT (IDENTIFIES SYSTEM FACTORS)

Claim:

Because:

Therefore:

The structural argument is more useful for prevention policy because: _____

Question B.3 — Prevention Proposal

3 marks

Propose ONE specific, evidence-based prevention strategy that directly addresses a structural barrier for People from multicultural backgrounds. Your proposal must include: what it is, who delivers it, how it addresses a structural barrier, and a measurable success metric.

MY PREVENTION PROPOSAL

Strategy:

Who delivers it:

Structural barrier addressed:

Success metric:

ACTIVITY C

Global Comparison — Australia vs Two Other Countries

⏱ ~15 min | VC2HP10P09 | Individual | Source: who.int · cdc.gov · royallifesaving.com.au

Use the data below and your own research to complete an international comparison of drowning rates and prevention approaches. Complete the data table, then answer the analysis questions.

Country	Approx. drowning rate (per 100,000)	Income category (World Bank)	Key prevention approach	Source URL
🇦🇺 Australia	1.2	High income		
🌍 Global average (WHO)	~4.0	Mixed		*who.int*
Country of your choice 1: Research and fill in				
Country of your choice 2: Research and fill in				

Question C.1 — Income and Drowning Risk

3 marks

The WHO reports that 92% of global drowning deaths occur in low- and middle-income countries. Using your data table and the WHO global average, explain the relationship between income level and drowning risk. Why is this relationship NOT simply explained by "low-income people take more risks"?

Source(s):

Question C.2 — What Can Australia Learn?

3 marks

Look at the prevention approaches of the two countries you researched. Identify ONE element from either country that Australia does NOT currently emphasise, and evaluate whether Australia should adopt it. Use the structured argument framework.

INTERNATIONAL ELEMENT TO CONSIDER

Element:

From (country):

Why it works there:

Should Australia adopt it? (Yes/No/Modified):

Justification:

Question C.3 — Reflection

2 marks

What does the international comparison tell you about the effectiveness of Australia's current water safety approach? Is Australia performing well relative to comparable high-income countries, or is there room for significant improvement?

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ACTIVITY D

Diving & Spinal Injury — Australian Data

~15 min | VC2HP10P02 · VC2HP10P08 | Individual

Sources: AIHW · RLSSA · Royal Perth Hospital

The following data relates to diving-related aquatic spinal injury in Australia. Unlike drowning data (which is reported annually), spinal injury data is compiled across multi-year studies. Use this data alongside the Victorian Drowning Report data in Sections 1–3.

10%
of all spinal cord injuries in Australia caused by unsafe diving
Source: AIHW

20%
of quadriplegia cases in Australia result from diving injuries
Source: AIHW

95%
of aquatic spinal injury patients are male
Source: Royal Perth Hospital spinal injury review

75%+
of aquatic spinal injury patients are under 30 years of age
Source: Royal Perth Hospital spinal injury review

66%
of aquatic spinal injury patients suffer permanent disability
Source: Royal Perth Hospital spinal injury review

10–14
years: highest incidence age group for diving-related spinal injuries
Source: AIHW

Post-COVID Youth Drowning Trend

Period	Age group	Drowning rate	Change
5 years pre-COVID	15–20 year olds	Baseline	—
5 years post-COVID	15–20 year olds	+34% above baseline	↑ 34%

Source: Royal Life Saving Society Australia (RLSSA) research. aihw.gov.au · royallifesaving.com.au

Prevention context: The core Australian response to this data is the **"Feet First, First Time"** message — always enter unknown water feet-first, and check depth on every visit regardless of prior experience. Inland waterway locations (rivers, dams, lakes) carry the highest drowning risk of any environment in Australia and present additional hazards beyond diving injury: cold water shock, poor visibility, and remote rescue delay.

Q4.1 — Data Comparison

3 marks

The Victorian drowning rate for 2024–25 was 0.75 per 100,000 population. Diving spinal injuries affect a significantly different population profile. Compare the

demographic distribution of Victorian drowning victims with the demographic distribution of aquatic spinal injury patients. What does this comparison reveal about the limitations of using drowning data alone to understand the full scope of dangerous water behaviour among young Australians?

Q4.2 — Rate Calculation Challenge

3 marks

The data shows 95% of aquatic spinal injury patients are male. If we assume the overall aquatic spinal injury population is evenly distributed across the age 10–30 male population in Australia (approximately 3.2 million), and that there are approximately 100 new cases per year nationally: (a) Calculate the rate per 100,000 for males aged 10–30. (b) How does this compare to the Victorian drowning rate (0.75 per 100,000 all ages)? (c) What are the significant limitations of this calculation?

Q4.3 — Causal Mechanism

4 marks

The post-COVID data shows a 34% increase in drowning among 15–20 year olds. Propose and evaluate THREE possible explanations for this increase. For each explanation, identify: (a) what evidence would support it, and (b) what prevention response would follow if it were the primary cause.

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Advanced Water Safety — Data Analysis Activity Pack · Year 9–10 HPE · Victorian Curriculum F–10 Version 2.0 · VC2HP10P02, VC2HP10P08, VC2HP10P09

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